

Linear independence

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The columns of a matrix A are linearly independent if and only if the equation of the homogeneous system $A\mathbf{x} = \mathbf{0}$ has *only* the trivial solution (**nontrivial solution does NOT exist**)

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The columns of a matrix A are linearly dependent if and only if the equation of the homogeneous system $A\mathbf{x} = \mathbf{0}$ has **nontrivial solution**

EXAMPLE 2 Determine if the columns of the matrix $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ are linearly independent.

SOLUTION To study $A\mathbf{x} = \mathbf{0}$, row reduce the augmented matrix:

$$\begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & -1 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & -2 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 13 & 0 \end{bmatrix}$$

At this point, it is clear that there are three basic variables and no free variables. So the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, and the columns of A are linearly independent. ■

EXAMPLE 3 Determine if the following sets of vectors are linearly independent.

a. $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

b. $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

$A = [\mathbf{v}_1 \ \mathbf{v}_2] =$

Augmented matrix of
homogeneous system $A\mathbf{x}=\mathbf{0}$:

Echelon Form:

Does the nontrivial solution exist?
Hint: Look at basic and Free variables

Finally, are they linearly independent?
Hint: see this slide above

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The columns of a matrix A are **linearly independent** if and only if the equation of the homogeneous system $A\mathbf{x} = \mathbf{0}$ has *only* the trivial solution (nontrivial solution does not exist).

Linear dependence

The columns of a matrix A are **linearly dependent** if and only if the equation of the homogeneous system $A\mathbf{x} = \mathbf{0}$ has *nontrivial* solution.

Repeat for part (b)

A Set of One
Vector

Linearly
independent
If it is not the zero
vector

Linearly
dependent
If it is the zero vector

A Set of Two
Vectors

A set of two vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if at least one of the vectors is a multiple of the other. The set is linearly independent if and only if neither of the vectors is a multiple of the other.

A set of two or more vectors is linearly dependent if any of these cases is true

A Set of Two or
More Vectors

If and only if at least one
of the vectors in the set
is a linear combination
of the others.

If the set contains
more vectors than
there are entries in
each vector.

If the set contains
the zero vector.

FACT

If it is **given** that a set of two or more vectors is linearly dependent and the first vector is not **0**, then some vector in the set is a linear combination of the preceding vectors

e.g.: $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6, \mathbf{v}_7, \dots, \mathbf{v}_p$

nonzero

This vector is a linear combination of all preceding vectors

Example:

Is this set of vectors linearly independent

$$\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix}$$

Yes, the vectors $\mathbf{u}$ and $\mathbf{v}$ are linearly independent because neither vector is a multiple of the other

Example:

Is this set of vectors linearly independent $\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

No, the vectors are linearly dependent because there are three vectors in the set and there are only two entries in each vector

Notice, even though none of the vectors is a multiple of one of the other vectors

EXAMPLE 6 Determine by inspection if the given set is linearly dependent.

a. $\begin{bmatrix} 1 \\ 7 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$

b. $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 8 \end{bmatrix}$

c. $\begin{bmatrix} -2 \\ 4 \\ 6 \\ 10 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ -9 \\ 15 \end{bmatrix}$

EXAMPLE 6 Determine by inspection if the given set is linearly dependent.

a. $\begin{bmatrix} 1 \\ 7 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$ b. $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 8 \end{bmatrix}$ c. $\begin{bmatrix} -2 \\ 4 \\ 6 \\ 10 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ -9 \\ 15 \end{bmatrix}$

SOLUTION

- a. The set contains four vectors, each of which has only three entries. So the set is linearly dependent by Theorem 8.
- b. Theorem 8 does not apply here because the number of vectors does not exceed the number of entries in each vector. Since the zero vector is in the set, the set is linearly dependent by Theorem 9.
- c. Compare the corresponding entries of the two vectors. The second vector seems to be $-3/2$ times the first vector. This relation holds for the first three pairs of entries, but fails for the fourth pair. Thus neither of the vectors is a multiple of the other, and hence they are linearly independent. 